
3.5.7 Gene expression is controlled by a number of features.

Most of a cell's DNA is not translated

Totipotent cells are cells that can mature into any body cell. During development, totipotent cells translate only part of their DNA, resulting in cell specialisation.

In mature plants, many cells remain totipotent. They have the ability to develop *in vitro* into whole plants or into plant organs when given the correct conditions.

Only a few totipotent cells, called stem cells, remain in mature animals. These can be used in treating some genetic disorders.

Candidates should be able to

- interpret data relating to tissue culture of plants from samples of totipotent cells
 - evaluate the use of stem cells in treating human disorders.
-

Regulation of transcription and translation

Transcription of target genes is stimulated only when specific transcriptional factors move from the cytoplasm into the nucleus.

The effect of oestrogen on gene transcription.

Small interfering RNA (siRNA) as a short, double-strand of RNA that interferes with the expression of a specific gene.

Candidates should be able to

- interpret data provided from investigations into gene expression
 - interpret information relating to the use of oncogenes and tumour suppressor genes in the prevention, treatment and cure of cancer
 - evaluate the effect on diagnosis and treatment of disorders caused by hereditary mutations and those caused by acquired mutations.
-
-